

Quantum-Assisted Multi-Asset Portfolio Construction

Step 03

Data Pipeline and Financial Inputs

Constructing returns, covariance, priors, liquidity, and transaction costs

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1 The role of Step 3

An optimizer can only be as credible as its inputs. Step 3 converts a 50-ETF universe into the numerical objects required by the portfolio model: expected growth, income, covariance, correlation, liquidity, linear trading cost, nonlinear impact, asset classes, and factor loadings.

Two data branches were implemented:

- a historical branch based on adjusted market data; and
- a synthetic heavy-tailed factor branch for reproducible challenge runs.

The final project results use the synthetic branch. The historical branch remains available, but its data were not used to support the final numerical claims.

2 The 50-asset universe

The universe spans US, developed, and emerging equities; core, government, inflation-linked, and credit bonds; real estate; precious metals; commodities; real-asset sectors; cash; FX; alternatives; and preferred securities. The design is deliberately multi-asset so that risk reduction can come from economically different sleeves rather than only from many similar equity funds.

3 Historical-return construction

For adjusted price $P_{i,t}$, simple daily return is

$$r_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1. \quad (1)$$

Missing prices are not converted into artificial zero returns. Extreme observations may be winsorized by clipping each column to lower and upper empirical quantiles.

A recency-weighted mean uses exponentially declining weights. If observation age is a_t days and half-life is h ,

$$\tilde{w}_t = 2^{-a_t/h}, \quad w_t = \frac{\tilde{w}_t}{\sum_u \tilde{w}_u}, \quad \hat{\mu}_{\text{EWMA}} = \sum_t w_t r_t. \quad (2)$$

Annualization multiplies the daily mean by 252.

The raw adjusted-price CAGR is

$$\text{CAGR} = \left(\frac{P_T}{P_0} \right)^{1/Y} - 1, \quad (3)$$

where Y is elapsed calendar years. CAGR is retained as a diagnostic, not mechanically substituted for the optimized return estimate.

4 Shrinkage of expected returns

Historical expected total return is shrunk toward a ticker-level prior:

$$\hat{\mu}_i = w_e \hat{\mu}_{i,\text{EWMA}} + (1 - w_e) \mu_{i,\text{prior}}. \quad (4)$$

In the configuration, the empirical-return weight is 0.35. The decomposition is

$$\hat{g}_i = \hat{\mu}_i - \hat{y}_i, \quad (5)$$

where $\hat{y}_i$ is estimated income yield. Shrinkage limits the tendency of noisy recent returns to dominate a forward-looking optimizer.

5 Covariance estimation and positive semidefiniteness

The historical branch uses a Ledoit-Wolf shrinkage estimator on a common sample. If S is the sample covariance and T a structured target, the conceptual form is

$$\hat{\Sigma} = \delta T + (1 - \delta)S, \quad (6)$$

with δ estimated from the data. The annual covariance is 252 times the daily covariance.

Numerical eigendecomposition is used to enforce positive semidefiniteness:

$$A = U \text{diag}(\lambda_1, \dots, \lambda_N) U^\top, \quad (7)$$

then

$$A_{\text{PSD}} = U \text{diag}(\max\{\lambda_i, \varepsilon\}) U^\top. \quad (8)$$

This prevents a tiny negative eigenvalue caused by numerical noise from producing a non-convex risk term.

Correlation is derived from covariance:

$$\rho_{ij} = \frac{\Sigma_{ij}}{\sqrt{\Sigma_{ii}\Sigma_{jj}}}. \quad (9)$$

6 Synthetic heavy-tailed factor model

The reproducible final branch simulates

$$r_t = \frac{\mu}{252} + B f_t + \varepsilon_t. \quad (10)$$

Here B is the asset-by-factor loading matrix, f_t is a multivariate Student- t factor shock with 7 degrees of freedom, and ε_t is Student- t idiosyncratic noise. Heavy tails produce more extreme observations than a Gaussian model.

For asset i , target variance is split into factor and idiosyncratic parts:

$$\sigma_i^2 = b_i^\top \Sigma_f b_i + \sigma_{\varepsilon,i}^2. \quad (11)$$

If the initial factor component is too large, loadings are rescaled so that factor variance does not exceed 90% of target variance. The remaining idiosyncratic variance is floored to avoid nearly deterministic assets.

The synthetic return series contains 1,260 business-day simulation observations. Its dates begin in 2000 as labels for the simulated path; they should not be interpreted as historical 2000 market data or as the stated 2021–2026 historical window.

7 Ticker-specific prior overlays

Class priors are modified for selected tickers to express economically intended differences:

- SCHD receives a higher income prior;
- USMV receives a lower volatility prior;
- QQQ and VUG receive growth-oriented adjustments;
- IWM receives higher small-cap risk.

The overlay table verifies that the transformations move the intended quantities rather than merely attaching labels.

Ticker	Expected return	Income yield	Optimizer volatility	Overlay
SPY	0.0700	0.0132	0.1736	False
QQQ	0.0611	0.0075	0.2096	True
VUG	0.0679	0.0042	0.1818	True
IWM	0.0810	0.0207	0.2140	True
USMV	0.0670	0.0159	0.1251	True
SCHD	0.0784	0.0321	0.1639	True

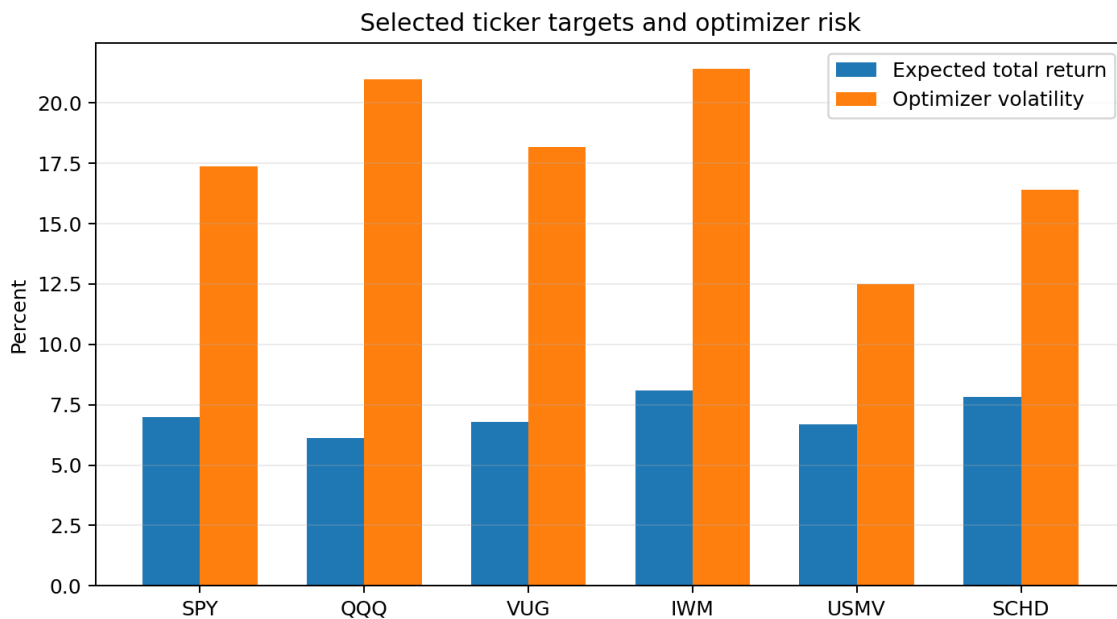


Figure 1: Selected final synthetic targets and optimizer volatilities.

8 Transaction-cost model

For portfolio value V , reference trade weight q , execution days D , and average daily dollar volume ADV_i , participation is

$$\pi_i = \frac{Vq}{D \text{ADV}_i}. \quad (12)$$

Daily volatility is $\sigma_{i,d} = \sigma_{i,a}/\sqrt{252}$. Square-root impact is

$$I_i = \eta \sigma_{i,d} \sqrt{\pi_i}. \quad (13)$$

The quadratic coefficient is calibrated so that a reference trade q has impact rate I_i :

$$\gamma_i = \frac{I_i}{q}. \quad (14)$$

The optimizer then uses approximately $\gamma_i(\Delta w_i)^2$. The base case uses a \$10 million portfolio, 3 execution days, and a 1% reference trade. The institutional sensitivity uses a larger portfolio and higher participation, producing greater modeled impact.

Ticker	Base participation	Institutional participation	Base bps	Institutional bps
SPY	0.000029	0.002210	1.383915	5.564788
QQQ	0.000026	0.001945	1.560433	6.297405
VUG	0.000016	0.001182	1.309859	4.512229
IWM	0.000072	0.005419	2.377371	10.450900
USMV	0.000048	0.003613	1.425575	5.278622
SCHD	0.000036	0.002681	1.551760	5.899990

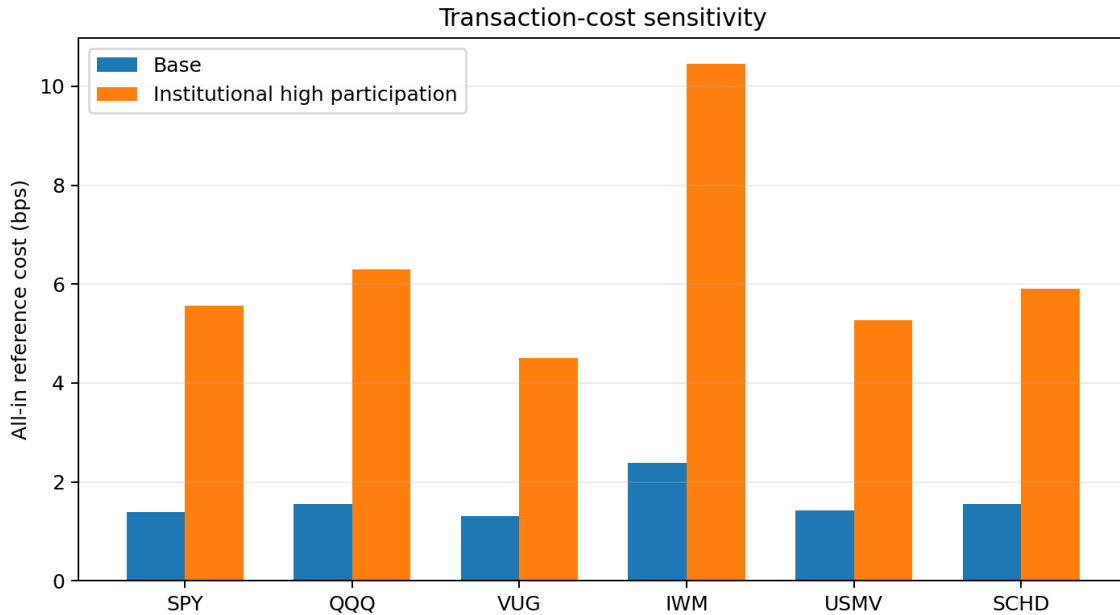


Figure 2: Institutional participation increases all-in reference trading costs.

9 Validation tests

Step 3 verifies:

- exactly 50 labeled assets;
- finite and symmetric covariance;
- nonnegative covariance eigenvalues within tolerance;
- unit diagonal and bounded correlations;
- finite, nonnegative costs and positive ADV;
- agreement between reported optimizer volatility and $\sqrt{\text{diag}(\Sigma)}$;
- higher participation and impact in the institutional sensitivity.

All final synthetic checks passed.

10 Outputs passed to the optimizer

The essential handoff is

$$\{g, y, \Sigma, c, \Lambda, \text{ADV}, \text{classes}, B\}. \quad (15)$$

The labels and ordering are preserved across every object. This prevents a serious silent error in which a covariance row for one ticker is applied to the weight of another.

11 Reproduction checklist

Use the 50-ticker universe, configuration values, synthetic seed 20260802, 1,260 observations, Student- t degrees of freedom 7, factor model, ticker overlays, PSD repair, and cost scenarios. Save all matrices with ticker labels and rerun the full validation dictionary before optimization.

12 Interpretation boundary

The synthetic branch is a controlled laboratory for optimization and algorithm validation. It is not historical backtesting. The parameters are economically structured, but their output is model evidence rather than a forecast guarantee.

A Data contract and why it matters

Step 3 converts informal market beliefs into typed numerical objects. Every later equation assumes that tickers appear in exactly the same order in the return vector, covariance matrix, cost table, scenario matrix, and constraint metadata. A silent ticker-order mismatch can produce a mathematically valid but economically meaningless portfolio. The data contract therefore stores one canonical ticker index and reindexes every table against it before optimization.

The minimum output schema is:

- daily return matrix $R \in \mathbb{R}^{T \times N}$;
- expected growth vector g and income vector y ;
- covariance Σ and correlation C ;
- class labels and optional factor loadings;
- asset caps, liquidity and average daily volume proxies;
- linear cost rates and quadratic impact coefficients;
- current portfolio weights; and
- scenario loss vectors and warning/hard thresholds.

B Historical branch: return construction

For adjusted prices $P_{t,i}$, simple daily return is

$$r_{t,i} = \frac{P_{t,i}}{P_{t-1,i}} - 1. \quad (16)$$

A missing price must not be forward-filled into a zero return, because that would artificially reduce volatility. The pipeline keeps the affected return missing and applies a common-observation rule when estimating covariance.

Annualized arithmetic mean and covariance are

$$\hat{\mu}_i = 252 \frac{1}{T} \sum_{t=1}^T r_{t,i}, \quad (17)$$

$$\hat{\Sigma} = 252 \frac{1}{T-1} (R - \bar{R})^\top (R - \bar{R}). \quad (18)$$

For a long sample, compound annual growth can be computed from adjusted-price endpoints:

$$\text{CAGR}_i = \left(\frac{P_{T,i}}{P_{0,i}} \right)^{1/Y} - 1, \quad (19)$$

where Y is elapsed years. The project keeps the historical branch available, but the final reported run uses synthetic inputs and states that scope explicitly.

C Synthetic factor model

Synthetic data is useful when the goal is to test the optimization architecture under controlled, reproducible assumptions. A standard factor representation is

$$r_t = \mu + B f_t + \varepsilon_t, \quad (20)$$

where $B \in \mathbb{R}^{N \times K}$ contains factor loadings, f_t is a vector of common shocks, and ε_t is idiosyncratic noise. If

$$\text{Cov}(f_t) = \Omega, \quad \text{Cov}(\varepsilon_t) = D, \quad (21)$$

then

$$\Sigma = B\Omega B^\top + D. \quad (22)$$

This construction is positive semidefinite by design when $\Omega \succeq 0$ and $D \succeq 0$. The final pipeline uses heavy-tailed shocks so extreme observations occur more often than under a Gaussian model. The random seed and simulation length are fixed for reproducibility.

The synthetic date index is a sequence of business-day labels, not historical market evidence. A reproduction should describe those labels as simulation dates rather than imply that the data were observed during those calendar years.

D Separating growth from income

Expected total return is decomposed as

$$\mu_i^{\text{total}} = g_i + y_i. \quad (23)$$

This permits a high-income asset and a high-growth asset to have similar total return but different investor roles. The pipeline checks this identity row by row. An optimizer can then reward $g^\top w$ and $y^\top w$ separately.

E Ticker-specific overlays

The base synthetic generator assigns parameters by asset class. Selected overlays then introduce economically interpretable differences:

- SCHD receives a higher income target;
- USMV receives a lower volatility target;
- QQQ and VUG receive growth-oriented adjustments;
- IWM receives higher small-cap risk.

An overlay is accepted only if a directional test passes. For example, the USMV volatility target must move downward relative to its pre-overlay class baseline. This is stronger than merely checking that code executed.

F Covariance and correlation validation

Given covariance Σ , volatility vector is

$$\sigma_i = \sqrt{\Sigma_{ii}}. \quad (24)$$

Correlation is

$$C_{ij} = \frac{\Sigma_{ij}}{\sigma_i \sigma_j}. \quad (25)$$

Validation checks include symmetry, finite entries, nonnegative diagonal, unit correlation diagonal, correlations within $[-1, 1]$ up to tolerance, and a minimum covariance eigenvalue no smaller than a small numerical tolerance. A small negative eigenvalue caused by floating-point rounding may be repaired by eigenvalue clipping, but the repair must be reported.

G Transaction-cost construction

Let s_i be a spread-like cost in basis points and q_i a trade participation rate. A stylized all-in cost is

$$c_i(q_i) = s_i + \eta_i \sqrt{q_i}, \quad (26)$$

where η_i controls nonlinear impact. The square-root form captures increasing cost with participation while allowing liquid assets to remain cheap. The project evaluates both a base participation case and a higher institutional-participation case. A directional audit requires

$$c_i(q_i^{\text{institutional}}) > c_i(q_i^{\text{base}}) \quad (27)$$

for applicable assets.

The optimizer separates linear transaction cost from quadratic impact. Therefore the pipeline saves enough information to reconstruct both rather than compressing them into one opaque number.

H Liquidity capacity

For portfolio value V , average daily dollar volume ADV_i , execution horizon D , and maximum participation ρ , the maximum trade weight is

$$q_i^{\max} = \frac{D\rho \text{ADV}_i}{V}. \quad (28)$$

This quantity enters Step 4 as an asset-level upper bound on $p_i + n_i$. Sensitivity to V , D , and ρ should be explicit because a portfolio feasible at one capital scale may not be feasible at another.

I Validation sequence

A robust data pipeline should stop before optimization if any of the following fail:

1. exactly 50 unique ticker identifiers are present;
2. all numerical vectors align to the canonical index;
3. expected total return equals growth plus income;
4. covariance and correlation checks pass;
5. optimizer volatility equals $\sqrt{\text{diag}(\Sigma)}$;
6. cost coefficients are finite and nonnegative;
7. the higher-participation cost case is more expensive;
8. current weights are finite, long-only, and sum to one; and
9. every class label and scenario vector has the correct dimension.

J Interpreting the final pipeline output

The final synthetic data are coherent with the intended asset roles, not predictions of market returns. The overlays move in the expected directions, covariance is positive semidefinite, and cost sensitivity behaves monotonically. These checks support the statement that later optimization results are internally consistent with the declared model. They do not validate the model against future realized returns.

K Complete reproduction protocol

1. Install the declared Python dependencies and set the seed.
2. Build the 50-asset universe and immutable ticker order.
3. Run either the historical download branch or the synthetic branch; never merge evidence labels between them.
4. Estimate or generate growth, income, covariance, and liquidity inputs.
5. Apply and directionally test the named ticker overlays.
6. Construct base and institutional cost scenarios.
7. Validate every matrix and vector.
8. Export CSV tables and the configuration before running Step 4.
9. Re-run the pipeline and compare dimensions, checksums, directional tests, and summary statistics.

Final implementation conventions

These notes align interpretation with the final executed code path and saved evidence without changing the reported numerical results.

- The model "current portfolio" is a synthetic/model reference incumbent constructed for the research experiment; it is not an observed Vanguard client portfolio.
- The final reported evidence uses the synthetic branch. Synthetic date indices are labels for simulated business-day observations, not historical market observations.
- The report configuration uses an empirical expected-return shrinkage weight of 0.35. Alternate historical-code defaults should be overridden by the recorded final configuration when reproducing the report.
- The 3-day horizon is used for the base market-impact calibration. Step 4 may use a separate 5-day liquidity/capacity window; these are different modeling quantities, not one shared execution-horizon parameter.